

Step 3 Validation Protocol: Dynamic Dimerisation/Phonon Coordinate

1 Purpose of the test

Step 2 showed that a scalar effective mixing,

$$V_{BD}^{\text{eff}} = V_0 + \lambda_\delta \delta + \lambda_C C_1, \quad (1)$$

can detect enhanced bright-dark mixing but cannot identify whether the enhancement comes from spin-Peierls dimerisation or local spin correlations. The purpose of Step 3 is to add a dynamic dimerisation/phonon coordinate and test whether this creates a spectroscopic signature that a static C_1 scalar control cannot reproduce.

2 Physical model

The orbital sector is the same minimal bright/dark model used in Steps 1 and 2. The Hilbert space contains a ground state $|g\rangle$, a dark excited state $|D\rangle$, a bright excited state $|B\rangle$, and a truncated harmonic coordinate with annihilation operator a . The phonon displacement is

$$Q = a + a^\dagger. \quad (2)$$

The Step 3 mixing Hamiltonian is

$$H_{\text{mix}} = V_{BD}^{\text{static}} L_{BD} + g_Q Q L_{BD}, \quad (3)$$

where

$$L_{BD} = |D\rangle\langle B| + |B\rangle\langle D| \quad (4)$$

and

$$V_{BD}^{\text{static}} = V_0 + \lambda_\delta \delta + \lambda_C C_1. \quad (5)$$

The new ingredient is not another scalar amplitude; it is the operator-valued term $g_Q Q L_{BD}$.

3 Expected results

At matched V_{BD}^{static} , the static dimerisation and static spin-correlation controls should remain equivalent, reproducing the degeneracy found in Step 2. The dynamic dimerisation/phonon case is expected to differ because it can mix phonon manifolds. Possible signatures include phonon sidebands, modified pathway amplitudes, altered peak positions, or changes in rephasing/unrephasing balance.

4 Numerical procedure

Run one scenario at a time by setting `STEP3_ACTIVE_CASE` or by editing `active_case_key` in the notebook. The default scenarios are:

Scenario	Purpose	Dynamic phonon term
<code>static_spin_correlation</code>	scalar C_1 control	off
<code>static_dimerisation</code>	scalar δ control	off
<code>dynamic_dimerisation_phonon</code>	dynamic lattice control	on

The third-order rephasing 1Q spectrum is generated using the same NQ protocol as in Steps 1 and 2. The frequency window is widened to leave room for phonon sideband structure.

5 Acceptance criteria

The first criterion is that the two static controls remain identical or nearly identical at matched V_{BD}^{static} . This confirms that the Step 2 degeneracy is reproduced.

The second criterion is that the dynamic dimerisation/phonon case differs from the static controls in at least one observable: cross-window area, cross/diagonal ratio, pathway-resolved amplitude, peak position, linewidth, or sideband intensity. If the dynamic case remains indistinguishable, then the chosen g_Q , phonon energy, Hilbert-space truncation, or observable is not sufficient to isolate lattice dynamics.

6 Requested figures and data

For each run, save:

- the rephasing real spectrum PDF;
- the pathway-resolved complex response arrays;
- the total rephasing, unrephasing, and 1Q components;
- text and JSON summaries containing V_{BD}^{static} , g_Q , ω_Q , δ , and C_1 .

Phase 2 analysis should compare finite-window integrals, cross/diagonal ratios, pairwise spectrum differences, and possible sideband windows offset by ω_Q .

7 Interpretation of possible failures

If the dynamic phonon case differs only by an overall amplitude, the observable may still be insensitive to true lattice dynamics. If the static controls do not match at equal V_{BD}^{static} , the model implementation should be checked before interpreting the dynamic case. If sidebands are absent, the phonon coupling may be too weak, the frequency window too narrow, or the truncated phonon basis too small.

8 Numerical results

The three Step 3 scenarios were generated at $\eta = 0.001$ with a 140×140 frequency grid, $n_{\text{bosons}} = 3$, and $\omega_Q = 0.035$ eV. The finite-window half-width used for the aggregate metrics is 0.025 eV. The two static controls were run at matched $V_{BD}^{\text{static}} = 0.02$.

Scenario	V_{BD}^{static}	g_Q	A_{cross}	$A_{\text{cross}}/A_{\text{diag}}$
Static spin correlation	0.0200	0	14.160315	0.142697
Static dimerisation	0.0200	0	14.160315	0.142697
Dynamic dimerisation phonon	0.0200	0.0100	11.761513	0.149235

The static dimerisation and static spin-correlation spectra are identical to numerical precision. Their relative maximum full-spectrum difference is 1.10×10^{-17} . This reproduces the Step 2 scalar degeneracy inside the larger phonon Hilbert space.

The dynamic dimerisation/phonon spectrum differs strongly from the static dimerisation control at the same V_{BD}^{static} . The relative maximum full-spectrum difference is 5.46×10^{-1} . The main cross-window area decreases from 14.160315 to 11.761513, while the cross/diagonal ratio increases from 0.142697 to 0.149235.

The sideband-window metric is qualitative because the finite windows can include tails from nearby resonances. Nevertheless, it also changes between the static and dynamic cases: the sideband/main-cross ratio changes from 2.21297 for the static controls to 2.41282 for the dynamic phonon case.

9 Validation outcome

The first criterion passes. The two static controls are identical at matched V_{BD}^{static} within relative maximum difference 1.10×10^{-17} . This confirms that simply embedding the scalar model in a larger phonon Hilbert space does not by itself break the degeneracy between δ and C_1 .

The second criterion also passes. Adding the dynamic term $g_Q Q L_{BD}$ produces a distinct non-linear spectrum relative to the static dimerisation control, with relative maximum full-spectrum difference 5.46×10^{-1} . Therefore an explicit dimerisation/phonon coordinate provides a viable route to separate lattice-driven spin-Peierls physics from a purely static local spin-correlation control.

10 Conclusions and next step

Step 3 resolves the main limitation of Step 2 at the model level. A scalar renormalisation of bright-dark mixing cannot identify the microscopic origin of the signal, but an operator-valued phonon coordinate can generate a distinct spectral response. This does not yet prove that a specific feature in CuGeO3 is phonon-driven, but it establishes a workable modelling criterion: lattice physics must enter through a dynamical or sector-specific degree of freedom, not only through a scalar V_{BD}^{eff} .

The next useful calculation is not to increase grid convergence immediately. Instead, the next step should scan the dynamical parameters one at a time:

- g_Q to determine the onset and approximate scaling of the dynamic phonon signature;
- ω_Q to check whether sideband displacement follows the phonon energy;

- rephasing versus unrephasing totals to see whether the dynamic coordinate changes pathway balance;
- real, imaginary, and absolute spectra to identify the most robust observable.

After that, the model should be connected to temperature and field trends by using $\delta(T, B)$ for the dynamic lattice channel and $C_1(T, B)$ for the static spin-correlation control.